

Phase 3 Project Design

FitBuddy AI Fitness Plan Generator

This phase describes how FitBuddy's browser experience, application services, AI generation and local storage work together. The design keeps the user flow direct while separating input validation, plan generation and persistence.

Design goals

- Provide one clear path from profile entry to a saved plan and follow-up feedback.
- Use a local-first architecture suitable for a student or prototype demonstration.
- Keep AI generation optional through the configured demo or live Gemini mode.
- Make safety boundaries and the source or mode of generated content visible.

Application components

Component	Responsibility
Browser interface	Collects profile details, displays the weekly plan and tip, and accepts feedback using Jinja templates and static assets.
FastAPI application	Receives browser and JSON requests, coordinates the plan workflow, and returns HTML or JSON.
Validation layer	Checks profile and feedback fields before use and defines accepted request and response shapes.
AI service	Builds workout and tip content using Gemini when configured, or the project's local/demo generation path.
Persistence layer	Uses SQLAlchemy with SQLite to store profiles, plans and revision information for local retrieval.

Primary request flow

- 1 The user submits a profile through the browser form or a JSON endpoint.
- 2 The application validates values and prepares the requested plan.
- 3 The AI service returns a workout plan and a separate general wellness tip.
- 4 The persistence layer saves the profile and plan; the interface displays the result and generation mode.
- 5 The user may submit feedback; the application revises the saved plan and returns the updated version.

Information and safety design

The application stores the information needed to associate a profile with a plan and its revisions. The original content and current revision should remain distinguishable. Model or mode labels help tell a live AI response from a built-in demo response.

Profile text and feedback are user data. They should not override safety instructions. Plan content should remain general, adapt conservatively to stated movement considerations, include recovery, and direct users with health concerns to qualified professionals.

Phase outcome

The design provides a small, testable service flow with a browser interface, JSON access, optional AI generation and local storage. Phase 4 schedules the work and defines review evidence.